

KAREEM SALAH

Robotics Engineer

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Professional Summary

Robotics Software Engineer experienced in developing autonomous systems, ROS-based navigation stacks, and robotic control pipelines, supported by a strong background in mechanical system design

Robotics & Autonomous Systems projects

EVER Egypt – Autonomous Vehicle Track

Autonomous Vehicle Software Engineer

08/2024 – 08/2025

- Built a full autonomous navigation stack using ROS, covering perception, planning, and control layers
- Implemented path tracking with Pure Pursuit and PID controllers for reliable trajectory following
- Developed and validated the simulation environment in CoppeliaSim before real-world deployment
- Integrated multi-sensor system including LiDAR, IMU, camera, and odometry; applied sensor fusion and filtering for robust localization
- Built an OpenCV-based vision pipeline for real-time object detection and classification (vehicles, cones, pedestrians)
- Debugged real-time software/hardware integration issues across sensing, computation, and actuation

Graduation Project – Mechatronics Dept., Assiut University

Bimanual Sign Language Communication System

10/2023 – 12/2024

- Developed a dual-arm robotic system using ROS and MoveIt for inverse kinematics and motion planning
- Built simulation and validation workflow using Gazebo and RViz; managed full integration before hardware deployment
- Trained and deployed YOLOv8 models (PyTorch) for real-time gesture recognition from live camera feed
- Integrated voice recognition and text-to-speech modules enabling bidirectional human-robot communication
- Led a 7-member cross-functional team through design, implementation, and integration phases

ASME XRC – Autonomous Robotics Competition

Autonomous Wheeled Robots – Competitive Systems

01/2022 – 05/2024

- Designed and programmed multiple autonomous robots: sumo combat, maze-solving, and puck collection platforms
- Implemented embedded control logic, sensor feedback loops, and navigation algorithms for each system
- Optimized real-time performance under competition constraints; achieved 1st internationally (ocean cleanup robot) and 2nd internationally (lunar lander)

Personal / R&D Projects

ESP IoT Firmware – Embedded Robotics Communication

- Developed firmware for ESP-based IoT nodes used for wireless robot control and telemetry
- Built OTA update and remote command interfaces enabling flexible multi-device deployment

Experience

Birr Machines

Design Automation Engineer – SOLIDWORKS API / C#

01/2026 – Present

- Co-initiated and currently developing a SOLIDWORKS API-based automation system in C# to auto-generate motor designs (parts, assemblies, and drawings) With integrated PDM system for version control
- Created and updated 3D CAD models, assemblies, and detailed manufacturing drawings.

Takamul Education

Robotics R&D Engineer

09/2025 – 12/2025

- Developed modular robotics control systems with voice command, OTA firmware updates, and multi-device interfaces
- Designed and prototyped educational robotic platforms with real-time sensor interaction and feedback
- Conducted reverse engineering on existing kits to improve reliability and performance of deployed platforms

NISBAU Egypt

Mechanical Design & Manufacturing Engineer

01/2025 – 09/2025

- Designed and adapted German concrete plant structures for local manufacture and site-specific assembly
- Conducted R&D on new plant configurations to improve efficiency and structural sustainability

Riyadh Dynamics

AMR Robotics Engineer

08/2024 – 12/2024

- Developed and deployed a full AMR navigation stack using ROS, covering localization, path planning, and motion control, with integrated PLC-based control

Grenium Startup

Co-Founder – Hydrogen Generator System

02/2022 – 05/2025

- Designed and implemented the control system for an HHO electrolyzer, integrating multiple sensors (pressure, temperature, current, gas concentration) along with valves for safe and efficient operation using STM32
- Developed monitoring and protection logic to regulate system performance and handle real-time faults

ASME – Assiut Student Section

Section President

05/2023 – 05/2024

- Led a team of 80+ members, organizing technical events, workshops, and competition participation
- Oversaw technical development programs and managed team participation in international robotics competitions

Skills

Robotics & Software: ROS / ROS2 • C++ / Python • SLAM / Localization • Motion Planning • Control Systems (PID, Pure Pursuit)

Simulation & Tools: Gazebo • CoppeliaSim • RViz / MoveIt

Perception: OpenCV • YOLOv8 / PyTorch • Sensor Fusion (LiDAR, IMU, Camera)

Embedded & Systems: ESP / AVR / STM32 • PLC • Real-time Systems • OTA Firmware

Mechanical & CAD: SOLIDWORKS (CSWP) • Autodesk Inventor • DFM & DFA • Vehicle Design

Education

Assiut University – Faculty of Engineering

B.Sc. Mechatronics Engineering

07/2019 – 10/2024

Assiut, Egypt

Training & Certificates

- SOLIDWORKS Champion – CSWA / CSWP: Certified SOLIDWORKS Professional
- HA Consulting Group: Automation, PLC and Classic Control
- Embedded system nano degree professional track

Key Achievements

ASME XRC International

Ranked 1st internationally – Autonomous Ocean Cleanup Robot; Ranked 2nd internationally – Autonomous Lunar Lander

EVER Egypt Autonomous Track

Ranked 2nd in Egypt

EVER Egypt Dynamic Track

Ranked 5th in Egypt

Robotics Challenge Egypt

Ranked 1st – Puck Collector Robot; Ranked 2nd – Sumo Robot

ASME Student Sections

ASME Membership Growth Award – Middle East & Africa (as chairman)

Languages

Arabic Native **English** Proficient **German** Intermediate

Find Me Online

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